YIFAN (STEVE) LYU

NLP & Trustworthy LLMs | PhD Applicant

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Google Scholar | GitHub

RESEARCH INTERESTS Social NLP · LLM Behavioral Evaluation · Structured NLP

I study how language models use linguistic and social context, combining controlled behavioral evaluation with structured system design to diagnose failure modes and build more reliable systems.

EDUCATION

Dalian University of Technology — Dalian, China

M.S. in Software Engineering, School of Software — Sep 2024 - expected Jun 2027

GPA: 3.63/4.00 | Average: 85.76/100

U.S. News 2026: #34 in Artificial Intelligence

Supervisor: Xiujuan Xu, Associate Professor

South-Central Minzu University — Wuhan, China

B.Eng. in Software Engineering — Sep 2020 - Jun 2024

GPA: 3.36/5.00 | Average: 84.15/100

Bachelor of Management in Financial Management

Thesis supervisor: Meng Shi, Associate Professor

PUBLICATIONS AND PREPRINTS

Peer-Reviewed Publications

1. Xinran Li, Xinze Che, **Yifan Lyu**, Zhiqi Huang, and Xiujuan Xu. "TCDA: Thread-Constrained Discourse-Aware Modeling for Conversational Sentiment Quadruple Analysis." *IJCAI-ECAI 2026 Main Track*, pp. 5810-5818. doi:10.24963/ijcai.2026/647.

   Models conversational sentiment quadruples by combining discourse-aware positional encoding with thread-constrained dialogue structure.

2. Xiujuan Xu, **Yifan Lyu**, Guanghui Li, and Zhehuan Zhao. "Time-Varying Propagation Graph Construction for Semantic Model Dissemination." *IEEE SmartIoT 2026*, pp. 400-401. doi:10.1109/SmartIoT70864.2026.00062.

3. Xiujuan Xu, Guanghui Li, **Yifan Lyu**, and Zhehuan Zhao. "Nonlinear Propagation Mechanisms of Semantic Models in Complex Topologies." *IEEE SmartIoT 2026*, pp. 402-403. doi:10.1109/SmartIoT70864.2026.00063.

Preprints

4. **Yifan Lyu**, Dianqing Lin, Xinran Li, Jiaqi Qiao, and Xiujuan Xu. "SPAR-Hate: Auditor-Guided Multi-Perspective Role Reasoning for Bilingual Hate Speech Parsing." arXiv preprint, 2026.

   Presents an auditor-guided multi-perspective role-reasoning framework for bilingual hate-speech parsing with evidence-constrained arbitration and sample-level reassembly.

5. **Yifan Lyu**, Xinran Li, Jiaqi Qiao, and Xiujuan Xu. "Signal or Spurious Cue? A Randomized Audit of Survey-Country Metadata in LLM Social Inference." arXiv preprint, 2026.

   Tests whether verified survey-country metadata improves LLM forecasts and whether randomly assigned country labels steer predictions.

6. Jiaqi Qiao, Xinran Li, **Yifan Lyu**, Xiujuan Xu, and Liu Yu. "Expert-Guided Multimodal Fusion for Unified Emotion and Sentiment Analysis." arXiv preprint, 2025.

7. Xinran Li, Xiaowei Zhao, Yubo Zhu, Zhiheng Zhang, Zhiqi Huang, Hongkun Song, Jinglu Hu, Xinze Che, **Yifan Lyu**, Yong Zhou, and Xiujuan Xu. "Extensible Multi-Granularity Fusion Network and Transferable Curriculum Learning for Aspect-based Sentiment Analysis." arXiv preprint, 2024.

RESEARCH EXPERIENCE

SPAR-Hate: Auditor-Guided Multi-Perspective Role Reasoning for Bilingual Hate Speech Parsing

Dalian University of Technology | Research Lead — May 2026 - Jul 2026

- Independently conceived and led the development of an auditor-guided multi-perspective role-reasoning framework for bilingual structured hate-speech parsing, combining local focus-unit decomposition, evidence-constrained arbitration, and sample-level reassembly.

- Designed the experimental protocol and bilingual data preparation; conducted baseline comparisons, component ablations, and bounded-arbitration diagnostics to assess the contribution of each reasoning component.
- With the same Qwen3-14B backbone, SPAR raised the reported average across main-track parsing metrics from 29.83 to 32.82 on ZH and from 29.59 to 37.51 on EN, relative to a zero-shot baseline.

Signal or Spurious Cue? A Randomized Audit of Survey-Country Metadata in LLM Social Inference

Dalian University of Technology | Research Lead Jun 2026 - Jul 2026

- Designed and executed a provenance-controlled behavioral audit separating verified metadata utility from directional uptake of randomly assigned country labels in LLM individual-response forecasts.
- Built an auditable inference pipeline spanning five LLMs, six countries, seven targets, and four metadata conditions.
- Constructed PROV-FORECAST with 14,400 paired probability distributions. On the separate 72-record panel, verified-country metadata reduced held-out Brier loss by 0.040 (95% CI [0.024, 0.056]); the estimated reduction in country-directed shifts after disclosing randomized-label provenance was 0.0003 (95% CI [-0.0157, 0.0166]).

TCDA: Thread-Constrained Discourse-Aware Modeling for Conversational Sentiment Quadruple Analysis

Dalian University of Technology | Research Collaborator Jan 2026 - Apr 2026

- Proposed and designed Discourse-Aware Rotary Position Embedding (D-RoPE), introducing topology-adaptive rotary encoding at token and utterance levels for conversational sentiment quadruple analysis.
- D-RoPE improved the MVQPN baseline from 35.68 to 37.52 Micro-F1 on ZH and from 35.62 to 36.42 on EN; the full TCDA model achieved 44.35 Micro-F1 on ZH and 39.69 on EN.

Expert-Guided Multimodal Fusion for Unified Emotion and Sentiment Analysis

Dalian University of Technology | Research Collaborator Oct 2025 - Jan 2026

- Contributed to prompt wrapping and pseudo-token injection; conducted LoRA fine-tuning experiments on MELD and MOSEI, plus four-backbone comparisons and modality ablations across MELD, CHERMA, MOSEI, and SIMS-V2.

Cross-Modal Emotion Understanding: A Transformer-GAT Approach for Dialogue Emotion Recognition

Dalian University of Technology | Research Collaborator Oct 2024 - Dec 2024

- Contributed EmoBERTa-based text feature extraction and modality ablations on IEMOCAP and MELD.

HONORS & AWARDS

- **First Prize**, National English Competition for College Students (NECCS), Non-English Major Category, 2022.
- **Academic Scholarship**, South-Central Minzu University, 2023 and 2024.
- **Honorable Mention**, Mathematical Contest in Modeling / Interdisciplinary Contest in Modeling (MCM/ICM).

TECHNICAL SKILLS

Programming & Frameworks: Python, PyTorch · LLM Engineering: vLLM, PEFT (LoRA/QLoRA), SFT, RLHF/GRPO, prompt design and debugging, API automation · Tools: Git/GitHub

RESEARCH METHODS & ANALYSIS

Experimental Design & Evaluation: hypothesis-driven experiments, controlled evaluation, behavioral auditing · Modeling & Analysis: dataset construction/alignment, structured prediction, ablation/error analysis · Statistical Inference: bootstrap CIs, permutation tests

TEACHING EXPERIENCE

Teaching Assistant, Artificial Intelligence and Compiler Technology | Sep 2024 - Jan 2025

Dalian University of Technology

- Graded coursework and guided hands-on compiler laboratories.

LANGUAGES

Mandarin Chinese: Native; PSC 93.1/100 (Grade 1-B).

English: IELTS Academic 8.0 (Listening 8.5; Reading 8.5; Writing 6.5; Speaking 7.5), January 2026.

Additional Languages: Korean (elementary); Spanish (beginner).